

Vortex Tech AI/ML Internship 2026

Week 1 Report – Data Cleaning & Exploratory Data Analysis (EDA)

Intern: Sarim Ahmed

Objective

To clean the IBM HR Employee Attrition dataset, perform exploratory data analysis, and prepare high-quality data for machine learning.

Tools & Technologies

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn

Work Completed

- Loaded and inspected the dataset.
- Verified rows, columns and data types.
- Checked missing values and duplicate records.
- Cleaned unnecessary data and prepared the dataset.
- Performed EDA using statistical summaries and visualizations.
- Analyzed relationships between employee attributes and attrition.
- Saved the cleaned dataset for future modeling.

Key Findings

Patterns indicated that variables such as overtime, income, job role, age and job satisfaction influence employee attrition. Visualizations made these relationships easier to understand and confirmed the importance of preprocessing before model development.

Skills Gained

Data Cleaning, Exploratory Data Analysis, Python Programming, Data Visualization, Feature Understanding, Data Preparation

Conclusion

Week 1 provided practical experience with cleaning and analyzing real-world HR data. The processed dataset was successfully prepared for machine learning implementation in Week 2.

Proof of Work:

```
Generate + Code + Markdown | Run All Restart Clear All Outputs | Jupyter Variables Outline ... Python 3.14.4
```

```
# Define project paths (best practice - makes code portable)
PROJECT_ROOT = Path.cwd().resolve()

# If the notebook is launched from the notebooks folder, move one level up
if not (PROJECT_ROOT / "data" / "raw").exists():
    PROJECT_ROOT = PROJECT_ROOT.parent

DATA_DIR = PROJECT_ROOT / "data" / "raw"

# Dataset filename (update if your filename is different)
DATASET_NAME = "WA_Fn-UseC_HR-Employee-Attrition.csv"

# Full path
data_path = DATA_DIR / DATASET_NAME

print(f"📁 Loading dataset from: {data_path}")

# Load the dataset
df = pd.read_csv(data_path)

print("✅ Dataset loaded successfully!")
print(f"Shape of dataset: {df.shape} (rows, columns)")
```

[6] Python

📁 Loading dataset from: C:\Users\Sarim Ahmed\NO_SYNC\week1_data_cleaning\data\raw\WA_Fn-UseC_HR-Employee-Attrition.csv
✅ Dataset loaded successfully!
Shape of dataset: (1470, 35) (rows, columns)

2. Initial Data Exploration

The dataset contains 1470 employee records with 35 features. No missing values and no duplicate rows were found.

```
Generate + Code + Markdown | Run All Restart Clear All Outputs | Jupyter Variables Outline ... Python 3.14.4
```

2. Initial Data Exploration

The dataset contains 1470 employee records with 35 features. No missing values and no duplicate rows were found.

```
print("First 5 rows of the dataset:")
display(df.head())

print("\nDataset Information:")
df.info()

# Show number of rows and columns
print(f"\nDataset Shape: {df.shape[0]} rows and {df.shape[1]} columns")
```

[7] Python

First 5 rows of the dataset:

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EmployeeCount	EmployeeNumber	...	Relation
0	41	Yes	Travel_Rarely	1102	Sales		1	2	Life Sciences	1	1	...
1	49	No	Travel_Frequently	279	Research & Development		8	1	Life Sciences	1	2	...
2	37	Yes	Travel_Rarely	1373	Research & Development		2	2	Other	1	4	...
3	33	No	Travel_Frequently	1392	Research & Development		3	4	Life Sciences	1	5	...
4	27	No	Travel_Rarely	591	Research & Development		2	1	Medical	1	7	...

5 rows × 35 columns

Cell 2 of 18

GenerateCodeMarkdownRun AllRestartClear All OutputsJupyter VariablesOutlinePython 3.14.4

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